

1811ICT/2807ICT Sample Exam T2-2026

⚠ This is a preview of the published version of the quiz.

Started: Oct 1 at 14:07

Quiz Instructions

This sample exam indicates the exam question types, but the real exam will have more questions.

Some of the questions will be auto-marked, therefore, please follow the instructions and pay attention to spelling. In particular, for data types, use the data types in Python exactly as they are defined, for example, use **int** rather than integer or Int, use **bool** rather than boolean or Bool, and **str** rather than string.

Concepts, functions and libraries that are not taught in this course are not allowed in the coding questions. Please pay attention to the indentation, do not use the HTML editor, just write directly in the textbox.

⋮

Question 1 2 pts

Is brisbane a valid variable name? Is sunshine-coast a valid variable name? Answer yes or no only.

brisbane sunshine-coast

substitution operator

sunshine_coast : yes

⋮

Question 2 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

✗ spaces, hyphens, punctuation, a leading digit, or a python keyword

What is the type and value of the following expression?

s in d
 string — dictionary — check if s is a key in the dictionary

Type Value

⋮

Question 3 2 pts

Given the following initialisations:

```

i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}

```

What is the type and value of the following expression?

j / i *division* *floating-point result*

Type

float

Value

0.5



Question 4 2 pts

Given the following initialisations:

```

i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}

```

What is the type and value of the following expression?

j // i *floor division* *2 // 4 = 0*

Type

int

Value

0



Question 5 2 pts

Given the following initialisations:

```

i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}

```

What is the type and value of the following expression?

f / j *3.5/2*

Type

float

Value

1.75



Question 6 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

True

p

or False

Type

bool

Value

True



Question 7 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

True

s.isalpha()

and not

p

False

Type

bool

Value

False



Question 8 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

```
i % j == j % i
4 % 2 = 0    2 % 4 = 2
0 == 2
```

Type

Value



Question 9 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

```
s[1]
```

Type

Value

`"p"`



Question 10 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

```
s[1:]
```

Type Value 

Question 11 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

step = -1

s[::-1]

Type Value 

Question 12 2 pts

Given the following initialisations:

```
i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}
```

What is the type and value of the following expression?

a[j][1] *a[2][1]* *a[2] selects the 3rd element (indexing starts at 0)*
||
(3, 9) *[1] selects the 2nd element of that tuple*

Type Value *a[2] = (3, 9)**a[2][1] = 9*

Question 13 2 pts

Given the following initialisations:

```

i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}

```

What is the type and value of the following expression?

→ = "apple"
`d[s]`

Type

int

Value

7



Question 14 2 pts

Given the following initialisations:

```

i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}

```

What is the type and value of the following expression?

→ dictionary
`j in d`

*j in d checks dictionary keys, not dictionary values!
 4 in d: False*

Type

bool

Value

False



Question 15 2 pts

Given the following initialisations:

```

i = 4
j = 2
f = 3.5
p = True
s = "apple"
a = [(1, 2), (2, 4), (3, 9)]
d = {'apple': 7, 'banana': 4}

```

What is the type and value of the following expression?

[y, x] for (x, y) in a

Type

list

Value

[(2, 1), (4, 2), (9, 3)]

(1, 2) → (2, 1)

x y

(2, 4) → (4, 2)

(3, 9) → (9, 3)

⋮

Question 16 1 pts

Which of the following statements are incorrect?

☒ Python variables are case sensitive

☒ lists are mutable, tuples are immutable

☒ set elements are unique

Elements in a set are unique and unordered

`__init__()` is not mandatory. A class can be defined without it, so omitting it does not cause a syntax error.

```
class Empty:
    pass
```

`__init__` is commonly used to initialise object attributes, but it is not required in every class.

☒ If you define a class in Python, you must define the constructor method `__init__()`. Otherwise there will be syntax error

⋮

Question 17 1 pts

Which of the following statements will return True?

☐ `bool("")` — False because the empty string is falsey

☒ `'12a'.upper() == '12A'` True

☐ `[1,2].extend((3,4)) == [1,2,(3,4)]` False

☐ `len((1.0, 1)) == 2` False
length is 1, not 2

⋮

Question 18 1 pts

In Python, which class is a superclass of all other classes?

object

Python's class hierarchy ultimately derives from object. A user-defined class therefore has object as an ancestor unless a different inheritance structure is specified.

⋮

Question 19 1 pts

How many times will the print statement in the following program execute?

```
for i in range(5):
    for j in range(i):
        print(j)
```

0 + 1 + 2 + 3 + 4 = 10

0, 1, 2, 3, 4

10

⋮

Question 20 2 pts

Show exactly what the following program will print.

```
for x in [1, 2, 4]:
    for y in [4, 2, 1]:
        if x != y:
            if y < x:
                print("apple")
            else:
                print("banana")
        else:
            print("cherry")
```

x=y (circled) *else*

x=1, y=4 → banana
y=2 → banana
y=1 → cherry
x=2 y=4 → banana
y=2 → cherry
y=1 → apple
x=4 y=4 → cherry
y=2 → apple
y=1 → apple

banana
banana
cherry
banana
cherry
apple
cherry
apple
apple

p



0 words

</>

+

-



Question 21 2 pts

Show exactly what the following program will print.

```
print([a, b] for a in "abc" for b in range(1, 3))
```

Edit View Insert Format Tools Table

*outer loop**list comprehension*

[('a', 1), ('a', 2), ('b', 1), ('b', 2), ('c', 1), ('c', 2)]

p



0 words



Question 22 5 pts

Write a program that reads whole numbers typed by the user until a zero is entered, then prints the list of positive numbers entered that can be divided by 3, all on the same line, separated by commas.

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12pt ▾ Paragraph ▾ | :

```

values = [ ]
while True:
    number = int(input(" "))
    if number == 0:
        break
    if number > 0 and number % 3 == 0:
        values.append(str(number))
print(", ".join(values))

```

not professional
set a flag

p



0 words



Question 23 5 pts

Write a function that, given a list of (product_name, quantity_sold) pairs, returns a dictionary that uses the product_name as key, and the total quantity_sold as values. Print the products in descending order of total quantities sold.

Example list: [('apple', 2), ('banana', 3), ('avocado', 1), ('banana', 5), ('apple', 2), ('bread', 1)]

Assume the function argument is valid. Do not handle erroneous arguments.

Edit View Insert Format Tools Table

12pt ▾ Paragraph ▾ | :

```
def total_quantities(pairs)
```

```
    totals = { }
```

```
    for product, quantity in pairs:
```

```
        if product not in totals:
```

```
            totals[product] = 0
```

```
            totals[product] += quantity
```

```
    return totals
```

```
pairs = [('apple', 2), ('banana', 3), ('avocado', 1),
```

```
        ('banana', 5), ('apple', 2), ('bread', 1)]
```

```
totals = total_quantities(pairs)
```

```
for product in sorted(totals, key=totals.get,
```

```
    reverse=True):
```

```
    print(product, totals[product])
```



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+

-



Question 24 8 pts

The file scores.txt contains the results for a series of rounds in a football competition. Lines in the file are either the start of a new round or the result of a match. A small example file might look like:

```
Round 1
Arsenal 0 0 Brighton & Hove Albion
Liverpool 2 1 West Ham United
Tottenham Hotspur 3 0 Manchester City
Round 2
Brighton & Hove Albion 0 2 West Ham United
Liverpool 3 1 Tottenham Hotspur
Manchester City 4 2 Arsenal
```

Write a program that reads scores.txt and prints the teams ranked by the number of goals, for example:

```
5 Liverpool
4 Tottenham Hotspur
4 Manchester City
3 West Ham United
2 Arsenal
0 Brighton & Hove Albion
```

Edit View Insert Format Tools Table

12pt Paragraph

```
goals = { }
```

```
with open('scores.txt', 'r') as file:
```

```
    for line in file:
```

```
        parts = line.split()
```

```
        if parts[0] == "Round":
```

```
            continue
```

```
            i = 0
```

```
            while not parts[i].isdigit():
```

```
                i += 1
```

```
            team1 = " ".join(parts[:i])
```

```
            score1 = int(parts[i])
```

```
            score2 = int(parts[i+1])
```

p



0 words



Question 25 4 pts

Explain why you wrote your code that way for the previous question. **Describe the logic** of your approach and **explain how** it solves the problem. No pseudo code. Keep it under 100 words.

Edit View Insert Format Tools Table

12pt Paragraph

```
team2 = ".join(parts[i+2:])
```

```
if team1 not in goals:
```

```
goals[team1] = 0
```

```
if team2 not in goals:
```

```
goals[team2] = 0
```

```
goals[team1] += score1
```

```
goals[team2] += score2
```

```
for team in sorted(goals, key=goals.get, reverse=True):
    print(goals[team], team)
```

↑
key
number

I read each line of scores.txt and ignore lines beginning with "Round". For each match, I find the first numeric field to separate the first team name from the two scores and the second team name. I store each team's total goals in a dictionary, adding the goals from every match. After processing the file, I sort the team names by their dictionary values in descending order and print each team with its total.

p



0 words



Question 26 10 pts

A class that models a GoCard account requires methods to:

- set up an account with an initial balance;
- adjust the balance when a trip is taken that costs a given fare;
- adjust the balance when the account is topped up;
- return the average fare for all the trips.

(a) Implement this class.

(b) Write statements that: create a new account with an initial balance of \$100.00; record a trip costing \$3.50; top up the account with \$20.00; and print the average fare.

(c) Create a subclass of your initial class that:

- sets up an account with an initial balance and a concession rate as a percentage for passengers who are granted concession fares, for example students; and
- applies the concession to all trips.

Edit View Insert Format Tools Table

12pt Paragraph

class GoCard:

def __init__(self, balance):

self.balance = balance

self.fares = []

def trip(self, fare):

self.balance -= fare

self.fares.append(fare)

def top-up(self, amount):

self.balance += amount

def average_fare(self):

if len(self.fares) == 0:

return 0

return sum(self.fares) / len(self.fares)

average

Saved at 14:08

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account = GoCard(100.00)

account.trip(3.50)

account.top-up(20.00)

print(account.average_fare())

class ConcessionGoCard(GoCard):

def __init__(self, balance, concession_rate):

super().__init__(balance)

self.concession_rate = concession_rate

def trip(self, fare):

concession_fare = fare * (1 - self.concession_rate / 100)

super().trip(concession_fare)